

VISEU, Portugal

WMO: 085600

Lat: 40.7150N

Lon: 7.8958W

Elev: 645

StdP: 93.81

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.3	0.8	-8.6	2.0	5.1	-6.1	2.4	5.4	11.1	4.0	10.1	5.0	5.1	70	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.7	33.0	18.8	31.1	18.2	29.4	17.8	20.5	28.9	19.7	28.0	19.0	26.9	3.1	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	13.9	22.9	17.1	13.2	21.9	16.4	12.6	21.0	61.8	28.8	59.0	28.2	56.3	26.9	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.9	8.6	7.6	-2.3	36.1	1.6	1.6	-3.4	37.3	-4.4	38.3	-5.2	39.2	-6.4	40.4		
			-3.9	22.3	1.6	1.1	-5.1	23.0	-6.0	23.7	-6.9	24.3	-8.1	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	7.2	8.0	10.1	11.5	14.7	18.4	20.7	21.4	19.1	14.9	9.7	7.9
	DBStd	6.17	2.66	3.20	3.58	4.03	4.09	4.12	3.90	3.77	3.69	3.72	3.02	2.50
	HDD10.0	359	93	71	44	27	4	0	0	0	0	4	41	74
	HDD18.3	2044	346	288	256	209	128	50	19	11	34	119	259	325
	CDD10.0	1694	5	16	47	72	149	252	331	353	272	156	32	8
	CDD18.3	338	0	0	0	4	15	52	92	106	57	13	0	0
	CDH23.3	3125	0	0	2	21	131	498	922	1041	440	71	0	0
	CDH26.7	1129	0	0	0	0	27	163	369	420	135	16	0	0
Wind	WSAvg	3.5	3.5	3.7	3.9	3.7	3.4	3.2	3.1	3.3	3.2	3.4	3.7	3.7
Precipitation	PrecAvg	1100	164	106	92	104	82	31	14	21	58	145	137	147
	PrecMax	1612	450	337	402	302	220	101	59	95	160	310	411	539
	PrecMin	648	0	0	0	30	3	0	1	0	2	23	0	4
	PrecStd	282	122	81	103	73	58	25	14	23	48	80	99	127
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	18.9	22.7	25.7	29.6	33.0	35.0	35.7	33.0	28.9	19.9	16.2
		MCWB	10.2	11.1	11.8	14.8	17.6	19.0	19.1	18.8	18.3	14.7	12.7	11.1
	2%	DB	13.9	16.7	20.3	23.2	26.5	30.4	32.8	33.2	30.1	25.1	17.4	13.9
		MCWB	9.0	10.1	11.0	13.7	16.4	18.4	18.9	18.6	16.9	14.8	12.4	9.9
	5%	DB	12.1	14.7	18.4	21.1	24.5	28.5	30.7	31.1	28.0	22.5	15.4	12.5
		MCWB	9.1	9.3	10.5	13.0	15.3	18.0	18.5	18.1	16.3	14.2	12.0	9.9
	10%	DB	10.8	12.8	16.2	18.5	22.3	26.3	28.5	29.0	25.9	20.4	14.0	11.4
		MCWB	9.0	8.7	9.8	11.9	14.5	17.5	18.3	17.7	15.9	13.6	11.3	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	12.6	13.5	16.2	19.1	21.4	21.5	21.3	19.6	17.9	15.5	12.8
		MCDB	13.2	16.2	18.4	23.2	27.0	28.4	29.7	30.4	28.8	22.2	16.6	13.7
	2%	WB	11.0	11.5	12.4	14.7	17.6	20.1	20.5	20.2	18.6	16.9	14.1	11.7
		MCDB	11.8	14.3	17.0	21.3	24.1	27.6	29.1	28.9	26.5	20.6	15.8	12.6
	5%	WB	10.3	10.4	11.5	13.5	16.3	19.1	19.7	19.3	17.7	16.0	13.0	10.8
		MCDB	11.1	12.9	15.6	19.4	22.7	26.2	28.1	27.9	24.8	19.1	14.5	11.7
	10%	WB	9.4	9.5	10.7	12.3	15.3	18.1	18.8	18.6	16.8	15.1	11.9	10.0
		MCDB	10.5	11.8	14.4	17.4	20.9	24.9	26.6	26.7	23.1	18.2	13.1	10.9
Mean Daily Temperature Range	5% DB	MDBR	6.1	7.4	8.1	8.6	9.9	11.2	12.7	12.7	10.6	7.6	6.2	5.9
		MCDBR	7.9	9.6	11.0	11.9	12.6	13.5	14.3	14.4	13.0	10.8	8.0	7.2
	5% WB	MCWBR	5.2	5.4	5.8	5.8	6.4	6.6	6.2	5.8	5.4	5.0	4.8	4.9
		MCWBR	5.5	7.8	9.0	11.0	11.7	12.9	13.9	13.5	11.7	8.0	6.0	5.5
Clear-Sky Solar Irradiance	taub	taub	0.298	0.310	0.341	0.344	0.351	0.354	0.344	0.358	0.346	0.334	0.308	0.296
		taud	2.549	2.508	2.431	2.427	2.427	2.431	2.441	2.396	2.459	2.517	2.548	2.553
	Ebn at Noon	Ebn at Noon	872	910	912	929	924	918	925	903	894	869	851	845
		Edh at Noon	71	86	103	111	114	113	111	113	99	84	71	66
All-Sky Solar Radiation	RadAvg	RadAvg	1.94	2.94	4.01	5.14	6.35	7.16	7.47	6.67	5.12	3.28	2.15	1.74
	RadStd	RadStd	0.31	0.43	0.59	0.64	0.65	0.45	0.29	0.31	0.39	0.39	0.38	0.26

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (16 neighbors)	+0.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+50

Nomenclature: See separate page